

Reference database of Raman spectra of biological molecules

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Raman spectra of biological materials are very complex, because they consist of signals from all molecules present in cells. In order to obtain chemical information from these spectra, it is necessary to know the Raman patterns of the possible components of a cell. In this paper, we present a collection of Raman spectra of biomolecules that can serve as references for the interpretation of Raman spectra of biological materials. We included the most important components present in a cell: (1) DNA and RNA bases (adenine, cytosine, guanine, thymine and uracil), (2) amino acids (glycine, L-alanine, L-valine, L-serine, L-glutamic acid, L-arginine, L-phenylalanine, L-tyrosine, L-tryptophan, L-histidine, L-proline), (3) fatty acids and fats (lauric acid, myristic acid, palmitic acid, stearic acid, 12-methyltetradecanoic acid, 13-methylmyristic acid, 14-methylpentadecanoic acid, 14-methylhexadecanoic acid, 15-methylpalmitic acid, oleic acid, vaccenic acid, glycerol, triolein, trilinolein, trilinolenin), (4) saccharides (β -D-glucose, lactose, cellulose, D-(+)-dextrose, D-(+)-trehalose, amylose, amylopectine, D-(+)-mannose, D-(+)-fucose, D-(–)-arabinose, D-(+)-xylose, D-(–)-fructose, D-(+)-galactosamine, N-acetyl-D-glucosamine, chitin), (5) primary metabolites (citric acid, succinic acid, fumarate, malic acid, pyruvate, phosphoenolpyruvate, coenzyme A, acetyl coenzyme A, acetoacetate, D-fructose-6-phosphate) and (6) others (β -carotene, ascorbic acid, riboflavin, glutathione). Examples of Raman spectra of bacteria and fungal spores are shown, together with band assignments to the reference products. Copyright © 2007 John Wiley & Sons, Ltd.

KEYWORDS: Raman spectroscopy; biology; biomolecules; reference Raman spectra

INTRODUCTION

Raman spectroscopy is a nondestructive, fast analysis technique that requires only small sample volumes and nearly no interference of water occurs. The bands in Raman spectra are well resolved compared to similar spectroscopic techniques, such as infrared spectroscopy. Therefore, it finds its application in a lot of research domains in chemistry, such as art analysis,¹ analysis of historical objects,² catalyst research,³ etc. Over the past few years, its possibilities in biology and microbiology have been explored. As mentioned by Carey,⁴ improvements have been made concerning problems such as low sensitivity, fluorescence rejection and data treatment, and thus the application of Raman spectroscopy became possible in these research fields.

Part of the research performed in these domains focuses on the identification of organisms such as bacteria, fungi and yeasts. Several of these studies have been successful in identifying species,^{5,6} sometimes even at the strain^{7–9} level.

For this purpose, spectra are treated as mathematical data and only little or no attention is given to the interpretation of the spectra. Despite a lot of (bio)chemical information being present in these spectra, it is hard to retrieve it because of the complex nature of Raman spectra of biological materials. Indeed, it is advantageous that the spectra contain contributions of all (complex) biomolecules from whole cells, but it is very hard to analyze them. Several research groups worked on the interpretation of Raman spectra of biological samples. For example, some Raman bands in bacterial spectra have already been assigned to simple biomolecules and groups of molecules, such as nucleic acids and proteins.¹⁰ In mycology, De Gussem *et al.*¹¹ analyzed *Lactarius* spore composition and Edwards *et al.*¹² studied lichens. Despite the progress in this field, there is still a lot of information included in these spectra that remains unused. Some authors have tried to avoid the complexity of the spectra by enhancing the contribution of specific molecules, e.g. by using surface enhanced Raman spectroscopy (SERS) or resonance Raman (RR) spectroscopy.^{13,14} This is an interesting approach when only one or a limited number of biomolecules are of interest. However, when an overview of the entire cell composition is

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needed, one has to rely to general Raman spectroscopy. A first step towards the interpretation of these complex spectra is the knowledge of the Raman band positions caused by the cell components. Lippert and Peticolas¹⁵ and Weng *et al.*¹⁶ have already presented Raman spectra of some fatty acids and fats. Surface enhanced Raman spectra of saccharides were recorded by Mrozek and Weaver,¹⁷ while Raman spectra of different types of carotenoids were described by Weesie *et al.*¹⁸

In this paper, we give an overview of Raman spectra of biomolecules. These spectra can serve as reference spectra for the interpretation of Raman spectra of biological materials. This database contains the most important building blocks of biomolecules present in a cell. It provides the basic information necessary to check the presence of biomolecules in biological materials, to monitor metabolism through time or to track changes induced by the environment. Finally, this approach is illustrated by the assignment of several bands in Raman spectra of bacteria and fungal spores.

EXPERIMENTAL

Raman instrumentation

Raman spectra were recorded with a Kaiser System Hololab 5000R modular Raman microspectrometer. A diode laser of 785 nm (Toptica Photonics AG) was focused through an 100x objective of the microscope (Leica), which resulted in a laser power of approximately 45 mW on the sample. The scattered light is transferred through a confocal 15 µm aperture pinhole and a collection fiber to the spectrograph, where it was detected by a back illuminated deep depletion Pelletier cooled (−70 °C) CCD detector (Andor). The Raman

signal was collected in the spectral interval of 150 cm^{−1} until 3500 cm^{−1}, but only the region from 365 to 1800 cm^{−1} was used. The acquisition time for all spectra was 60 s and the spectral resolution was approximately 4 cm^{−1}.

Spectral calibration was performed in MATLAB as described by Hutsebaut *et al.*,¹⁹ which includes (1) absolute wavelength calibration with a neon lamp, (2) intensity calibration with a tungsten lamp, (3) relative wavelength calibration with seven reference products, (4) correction for dark noise of the detector and (5) correction for the spectral contribution of optical components.

Reference products

Table 1 lists all 61 reference products along with their supplier and technical information. The products were divided into six groups: DNA and RNA bases, amino acids, fatty acids and fats, saccharides, primary metabolites and others. For analysis, the reference products were transferred to a CaF₂ slide. For each reference product, different focus points (2–4) were chosen, where each five Raman spectra of 60 s were collected. These calibrated spectra were treated by extended multiplicative scattering correction (EMSC),²⁰ using the EMSC toolbox, developed by H. Martens.²¹ In particular, dataset 103 was applied, which represents a correction for physical interferences. Afterwards, the spikes were removed, followed by the calculation of the mean spectrum for each reference product.

RESULTS AND DISCUSSION

Figures 1–12 show Raman spectra of DNA and RNA bases, amino acids, fatty acids and fats, saccharides, primary

Table 1. List of the reference products along with their supplier and technical information

Name	Supplier	Specifications
<i>DNA and RNA bases</i>		
Adenine	Acros organics	99%
Cytosine	Sigma-Aldrich	
Guanine	Sigma-Aldrich	98%
Thymine	Sigma-Aldrich	
Uracil	Sigma-Aldrich	Crystalline
<i>Amino acids</i>		
Glycine	Sigma-Aldrich	
L-Alanine	Sigma-Aldrich	Reagent grade, 98% (TLC)
L-Valine	Sigma-Aldrich	Reagent grade, 98% (TLC)
L-Serine	Sigma-Aldrich	ReagentPlus®, 99% (TLC)
L-Glutamate (monosodium salt hydrate)	Sigma-Aldrich	≥99% (TLC), powder
L-Arginine	Sigma-Aldrich	Free base
L-Phenylalanine	Sigma-Aldrich	
L-Tyrosine	Merck	For biochemistry
L-Tryptophan	Sigma-Aldrich	Reagent grade, ≥98%
L-Histidine	Sigma-Aldrich	ReagentPlus®, 99% (TLC)
L-Proline	Sigma-Aldrich	Biochimika, 99% (NT)

Table 1. (Continued)

Name	Supplier	Specifications
<i>Fatty acids and fats</i>		
Lauric acid	Sigma-Aldrich	98%
Myristic acid	Sigma-Aldrich	Sigma grade, 99–100%
Palmitic acid	Sigma-Aldrich	Free acid, Sigma grade
Stearic acid	Sigma-Aldrich	Free acid, grade I, ~99%
12-Methyltetradecanoic acid	Sigma-Aldrich	
13-Methylmyristic acid	Sigma-Aldrich	
14-Methylpentadecanoic acid	Sigma-Aldrich	
14-Methylhexadecanoic acid	Sigma-Aldrich	
15-Methylpalmitic acid	Sigma-Aldrich	
Oleic acid	Sigma-Aldrich	Reagent grade, 99%
cis-vaccenic acid	Sigma-Aldrich	
Glycerol	Sigma-Aldrich	BioChemika Ultra, anhydrous, ≥99.5% (GC)
Triolein	Sigma-Aldrich	(C18:1,-CIS-9) Sigma grade approx 99%
Trilinolein	Sigma-Aldrich	(C18:2,-CIS,CIS-9,12) approx 99%
Trilinolenin	Sigma-Aldrich	Purum ~98% (GC)
<i>Saccharides</i>		
β-D-Glucose	MP Biomedicals	99%
Lactose	DMV international (Veghel, the Netherlands)	Respitose SV003
Cellulose	MP Biomedicals	98%
D-(+)-Dextrose	MP Biomedicals	Anhydrous
D-(+)-Trehalose	Sigma-Aldrich	Biochimika ~99.5% (HPLC)
Amylose	MP Biomedicals	Essential free of amylopectin
Amylopectin	MP Biomedicals	
D-(+)-Mannose	Sigma-Aldrich	For microbiology
D-(+)-Fucose	Merck	
D-(+)-Arabinose	Merck	
D-(+)-Xylose	Merck	
D-(+)-Fructose	Sigma-Aldrich	99%
D-(+)-galactosamine hydrochloride	Sigma-Aldrich	Cell culture tested
N-Acetyl-D-glucosamine	Sigma Aldrich	≥99%, powder
Chitin	MP Biomedicals	Practical grade 90–95%
<i>Primary metabolites</i>		
Citric acid	Sigma-Aldrich	Anhydrous, 99%
Succinic acid	Merck	
Fumarate (sodium salt)	Merck	
Malic acid	Merck	
Pyruvate (sodium salt)	Sigma-Aldrich	Reagent plus TM, ≥99%
Phosphoenolpyruvate	Boehringer Mannheim	
Coenzyme A	Sigma-Aldrich	Free acid from yeast, hydrate
Acetyl coenzyme A	Sigma-Aldrich	
Acetoacetic acid (lithium salt)	Sigma-Aldrich	
D-Fructose-6-phosphate (disodium salt)	Sigma-Aldrich	
<i>Others</i>		
β-carotene	Sigma-Aldrich	Type II, synthetic, ≥95% (HPLC), crystalline
Ascorbic acid	Certa	Ph. Eur.
Riboflavin (sodium phosphate)	Certa	
Glutathione (reduced form)	Boehringer mannheim	

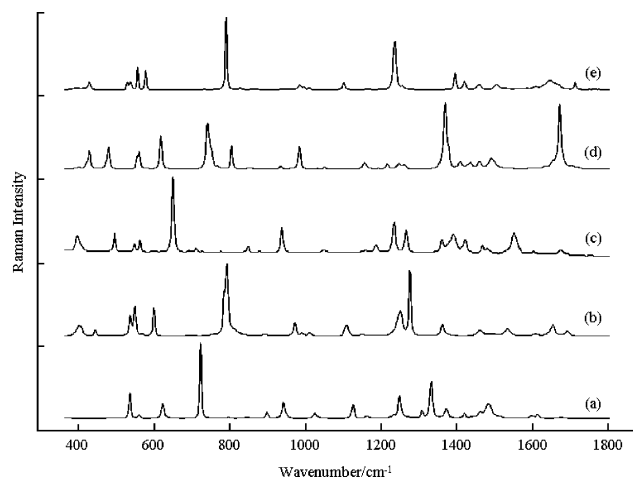


Figure 1. Raman spectra of DNA and RNA bases: (a) adenine, (b) cytosine, (c) guanine, (d) thymine, (e) uracil.

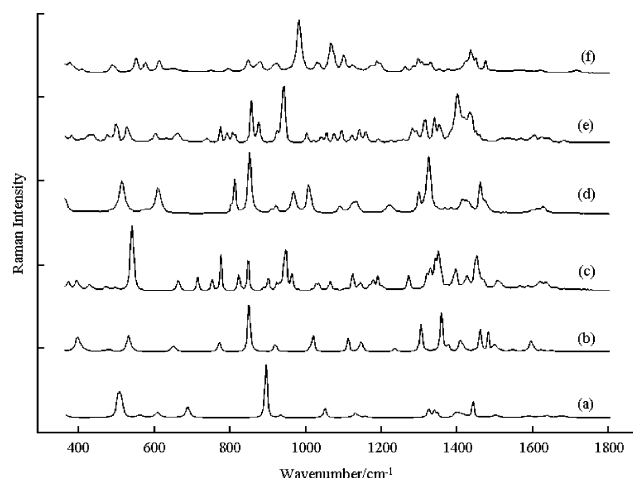


Figure 2. Raman spectra of amino acids with a non-cyclic R side chain: (a) glycine, (b) L-alanine, (c) L-valine, (d) L-serine, (e) L-glutamate, (f) L-arginine.

metabolites and others. The spectra of these groups can clearly be distinguished from each other. The spectra of DNA and RNA bases are dominated by ring breathing vibrations in the region from 600 until 800 cm^{-1} (Fig. 1). In the spectra of cells, not only the vibrations of these bases, but also symmetric $\nu(\text{PO}_2^-)$ stretches of the DNA and RNA backbone will be visible at *ca.* 1100 cm^{-1} . In general, the presence of proteins can be noticed by amide I and amide III bands at about 1300 and 1655 cm^{-1} . Raman bands of specific amino acids (Figs 2 and 3) are discussed below. Raman spectra of pure fatty acids and fats (Figs 4–6) were collected and can be distinguished from the other groups by typical bands at about 1300 and 1440 cm^{-1} . Spectra of saccharides (Figs 7–9) are characterized by groups of bands in the regions 1000 – 1200 cm^{-1} and 1300 – 1500 cm^{-1} . The spectra of the primary metabolites (Figs 10 and 11) are mutually very different.

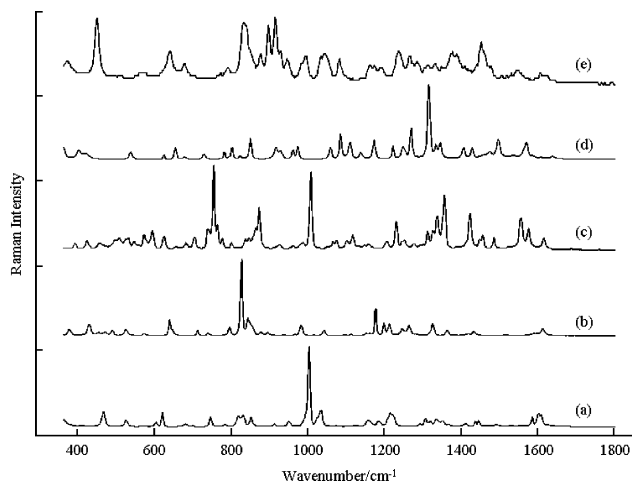


Figure 3. Raman spectra of amino acids with a cyclic R side chain: (a) L-phenylalanine, (b) L-tyrosine, (c) L-tryptophan, (d) L-histidine, (e) L-proline.

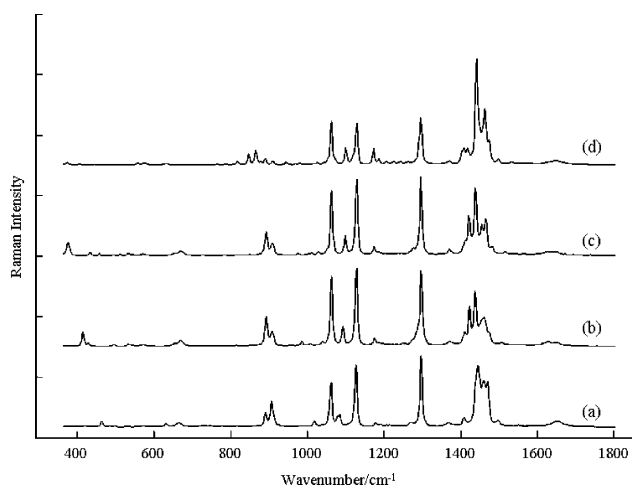


Figure 4. Raman spectra of saturated linear fatty acids: (a) lauric acid, (b) myristic acid, (c) palmitic acid, (d) stearic acid.

DNA and RNA bases

The Raman spectra of the five bases present in DNA and RNA are shown in Fig. 1. Each spectrum contains an intense band in the region from 600 to 800 cm^{-1} that can be assigned to ring breathing vibrations. The $\nu(\text{C}=\text{O})$ stretch vibrations cause only for thymine an intense band (1671 cm^{-1}). In the spectra of all five bases, almost no similarities can be noticed, so it is easy to distinguish between them. Even T and U do not show similarities while their chemical structure only differs in the substitution of a methyl group at the C5 position (Appendix). These five bases can be divided into purine bases (adenine and guanine) and pyrimidine bases (cytosine, thymine and uracil), but there are no bands observed that are characteristic for each group, as reported by Socrates.²²

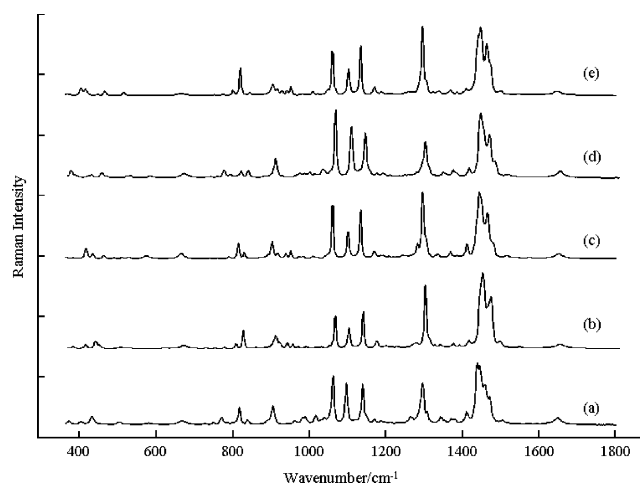


Figure 5. Raman spectra of saturated branched fatty acids: (a) 12-methyltetradecanoic acid (15Aiso), (b) 13-methylmyristic acid (15iso), (c) 14-methylpentadecanoic acid (16iso), (d) 14-methylhexadecanoic acid (17Aiso), (e) 15-methylpalmitic acid (17iso).

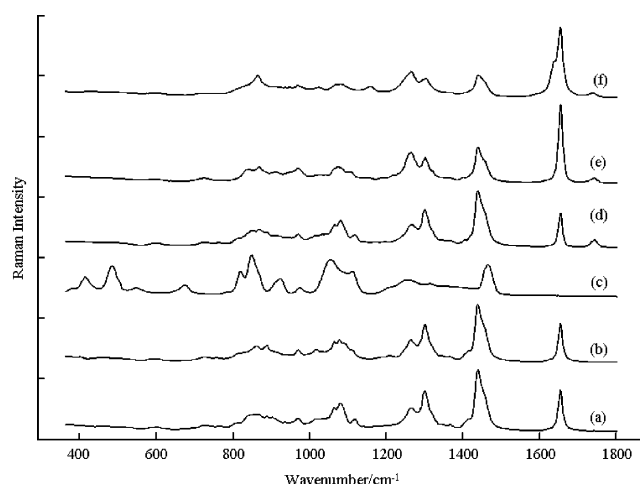


Figure 6. Raman spectra of the unsaturated fatty acids (a) oleic acid and (b) *cis*-vaccenic acid, of (c) glycerol and of the fats (d) triolein, (e) trilinolein and (f) trilinolenin.

Amino acids

Raman spectra of 11 amino acids with a variety of functional groups on the R side chains were recorded. In Fig. 2, Raman spectra of six amino acids without a ring structure on the R side chain are shown. Lin-Vien *et al.*²³ describe that several Raman bands for carboxylic acids and NH_2 groups are rather weak. This can explain why the amino acids, which have these two functionalities in common, show little similarities in their Raman spectra. In the Raman spectra of glycine (2a) and L-alanine (2b), which have small and simple R side chains, an intense band in the region between 850 and 900 cm^{-1} appears, which can be ascribed to the $\nu(\text{CNC})$ symmetric stretch of the amino group.²³ In amino

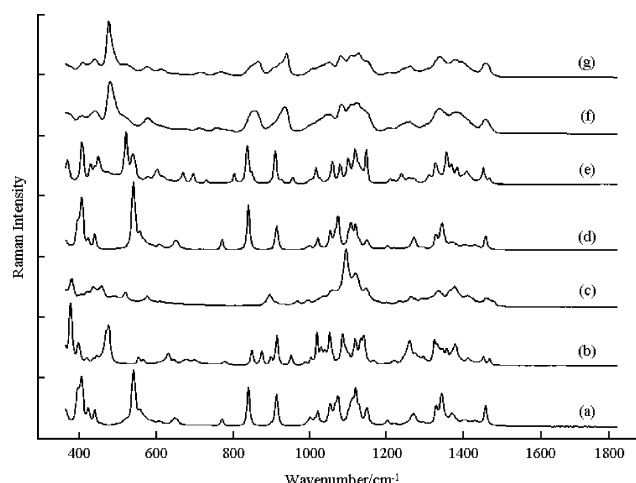


Figure 7. Raman spectra of saccharides: (a) β -D-glucose, (b) lactose, (c) cellulose, (d) D-(+)-dextrose, (e) D-(+)-trehalose, (f) amylose, (g) amylopectin.

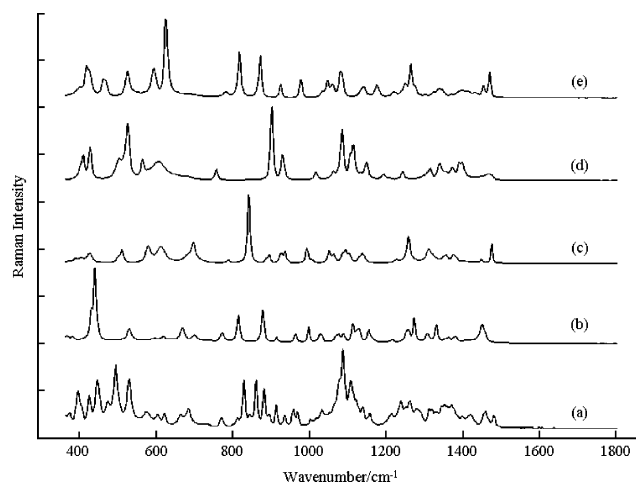


Figure 8. Raman spectra of saccharides: (a) D-(+)-mannose, (b) D-(+)-fructose, (c) D-(-)-arabinose, (d) D-(+)-xylose, (e) D-(-)-fructose.

acids with a more extended or complex R side chains, the influence of this R group on the position of the $\nu(\text{C}-\text{N})$ stretch band can be large, and therefore it is hard to assign a band to this vibration. For all amino acid spectra in Fig. 2, $\delta(\text{CH}_2)$ or $\delta(\text{CH}_3)$, deformations are observed in the region 1420–1500 cm^{-1} . For L-valine (2c), L-serine (2d) and L-glutamate (2e), some bands are observed originating from functional groups of the R side chain. In the spectrum of L-valine 2(c), $\delta(\text{CC}_3)$ deformation and tertiary $\delta(\text{CH})$ deformation of the isopropyl group give rise to bands at respectively 542 and 1351 cm^{-1} .²³ For L-serine (2d), the bands 813 and 853 cm^{-1} are located in the region where in phase $\nu(\text{C}-\text{C}-\text{O})$ stretches of primary alcohol functions as well as $\nu(\text{C}-\text{N})$ stretches of amines are expected. The symmetric $\nu(\text{CO}_2^-)$ stretch of L-glutamate (2e) can be observed at

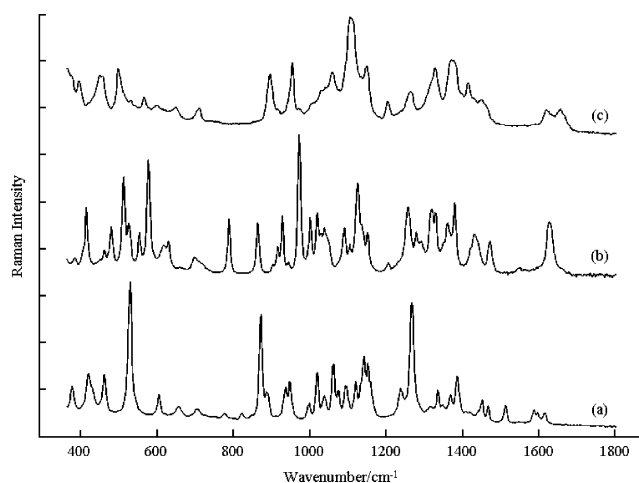


Figure 9. Raman spectra of nitrogen containing saccharides: (a) D-(+)-galactosamine, (b) *N*-acetyl-D-glucosamine, (c) chitin.

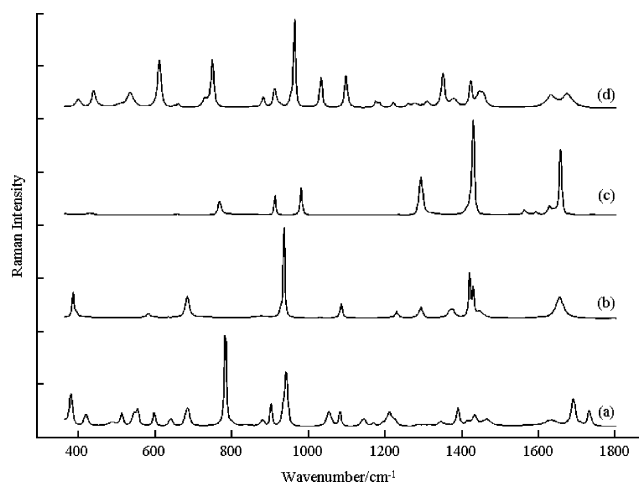


Figure 10. Raman spectra of components of the citric acid cycle: (a) citric acid, (b) succinic acid, (c) fumarate, (d) malic acid.

1401 cm^{-1} . Raman spectra of five amino acids of which the R side chain contains a ring structure are shown in Fig. 3. For L-phenylalanine (3a), L-tyrosine (3b) and L-tryptophan (3c), which all contain a benzene structure, the most intense Raman bands can be attributed to ring vibrations. Indeed, the intense bands at 1004 and 1009 cm^{-1} present in the spectra of respectively L-phenylalanine and L-tryptophan can be assigned to the trigonal ring breathing of the benzene ring. As L-tyrosine contains a *para*-substituted benzene ring, its ring breathing vibration is located at 828 cm^{-1} . The Raman bands at 1358 and 1423 cm^{-1} of L-tryptophan, are described as typical for indole rings by Socrates.²² For L-histidine and L-proline, it is hard to attribute the Raman bands to specific functional groups. Instead, quantum chemical calculations would be necessary in order to perform accurate band assignments.

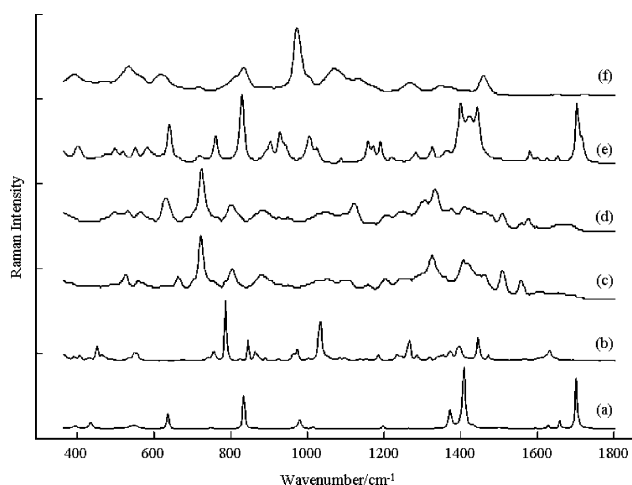


Figure 11. Raman spectra of primary metabolites related to the citric acid cycle: (a) pyruvate, (b) phosphoenolpyruvate, (c) coenzyme A, (d) acetyl coenzyme A, (e) acetoacetate, (f) D-fructose-6-phosphate.

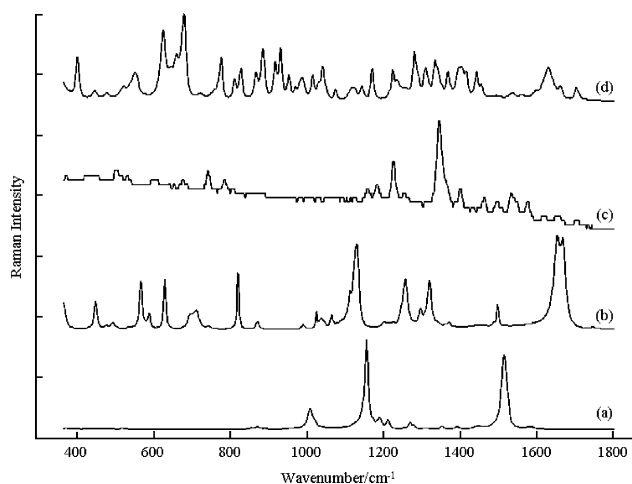


Figure 12. Raman spectra of (a) β -carotene, (b) ascorbic acid, (c) riboflavin, (d) glutathione.

Fatty acids and fats

The spectra of four unbranched saturated fatty acids are shown in Fig. 4. Characteristics for these fatty acids are the three Raman bands between 1050 and 1150 cm^{-1} , the band at 1296 cm^{-1} and a group of bands in the region between 1400 and 1500 cm^{-1} due to $\nu(\text{C}-\text{C})$ stretching vibrations, $\delta(\text{CH}_2)$ twist vibrations and $\delta(\text{CH}_3)$ or $\delta(\text{CH}_2)$ deformations, respectively. As the fatty acid chain elongates with two carbon atoms, several effects in the Raman spectra can be observed. Two shifts occur with increasing chain length: (1) the $\nu(\text{C}-\text{C})$ stretch vibration band at 1084 cm^{-1} in the spectrum of lauric acid (4a) shifts over 1092 cm^{-1} (myristic acid, 4b) and 1099 cm^{-1} (palmitic acid, 4c) to 1100 cm^{-1} for stearic acid (4d) and (2) the chain expansion vibration shifts from 464 cm^{-1} for lauric acid (4a) over 414 cm^{-1} (myristic

acid, 4b) to 375 cm^{-1} for palmitic acid (4c). The relative intensity of the bands at approximately 891 and 908 cm^{-1} , due to $\rho(\text{CH}_2)$ rocking vibrations, changes with the chain length.

In Fig. 5, spectra of saturated branched fatty acids are shown (full names together with their abbreviations are given in Table 2). Although the general profile of these Raman spectra is similar to the spectra of the linear fatty acids, there are some clear differences. Comparison of the Raman spectra

of palmitic acid (4c) and 16iso (5c) shows that the $\delta(\text{CH}_2)$ and $\delta(\text{CH}_3)$ deformations result in only two intense bands in the region from 1400 to 1500 cm^{-1} for branched fatty acids, while the Raman spectra of the linear fatty acids show four to five smaller bands in that region. The $\rho(\text{CH}_2)$ rocking doublet at 893 and 909 cm^{-1} in palmitic acid (4c) is decomposed into several bands in the spectrum of 16iso (5c) and some additional bands appear around 800 cm^{-1} . In the region of the chain expansion vibrations (below 470 cm^{-1}), more bands

Table 2. Raman bands and intensities for each reference product (very weak bands are not listed)

Product	Raman bands (in cm^{-1})	Fig
Adenine	536(m), 560(w), 623(m), 723(s), 898(w), 941(m), 1024(w), 1125(m), 1162(w), 1134(w), 1248(m), 1307(w), 1332(m), 1371(w), 1419(w), 1462(w), 1482(m), 1612(w)	1(a)
Cytosine	402(mw), 444(w), 537(m), 548(m), 568(w), 599(m), 792(s), 971(mw), 991(w), 1011(w), 1108(mw), 1250(m), 1275(s), 1362(mw), 1460(w), 1493(w), 1533(w), 1653(mw), 1690(w)	1(b)
Guanine	397(m), 496(m), 548(w), 563(m), 650(s), 711(w), 849(w), 937(m), 1048(w), 1158(w), 1187(w), 1234(m), 1266(m), 1360(m), 1391(m), 1422(m), 1468(w), 1479(w), 1550(m), 1602(w), 1674(w)	1(c)
Thymine	429(m), 479(m), 561(m), 617(s), 740(s), 767(w,sh), 804(m), 984(m), 1156(w), 1216(w), 1247(w), 1261(w), 1369(vs), 1408(w), 1435(w), 1459(w), 1490(m), 1655(w,sh), 1671(vs), 1702(w)	1(d)
Uracil	429(w), 529(w), 537(w), 556(m), 577(m), 790(vs), 984(w), 995(w), 1100(w), 1235(s), 1255(w), 1394(m), 1418(m), 1455(w), 1504(w), 1608(w), 1646(m), 1711(w)	1(e)
Glycine	504(s), 560(w), 606(m), 686(m), 894(vs), 932(w), 1049(m), 1129(m), 1156(w), 1323(m), 1338(m), 1346(m), 1398(m), 1441(m), 1499(w), 1589(w), 1627(w), 1634(w), 1674(w)	2(a)
L-Alanine	398(ms), 480(w), 532(ms), 652(mw), 773(m), 851(vs), 920(mw), 1021(m), 1112(m), 1146(m), 1236(w), 1305(s), 1359(vs), 1378(mw), 1409(m), 1462(s), 1482(s), 1499(m), 1596(m)	2(b)
L-Valine	374(mw), 396(mw), 429(mw), 472(w), 497(w), 542(vs), 664(mw), 715(m), 753(m), 776(s), 824(m), 849(s), 891(w,sh), 902(m), 923(mw,sh), 948(s), 964(m), 1029(mw), 1035(mw), 1066(mw), 1106(w), 1125(m), 1146(mw), 1179(mw), 1191(m), 1272(m), 1321(m,sh), 1330(m), 1343(s,sh), 1351(s), 1398(m), 1427(m), 1452(s), 1508(mw), 1567(w), 1587(w), 1619(mw), 1633(mw), 1660(w)	2(c)
L-Serine	515(m), 611(m), 813(m), 853(s), 910(w,sh), 922(w), 968(m), 1008(m), 1090(w), 1134(mw), 1220(mw), 1299(m), 1326(s), 1368(w), 1384(w), 1416(mw), 1426(mw,sh), 1462(m), 1599(w), 1613(w), 1628(w)	2(d)
L-Glutamate	382(mw), 429(mw), 477(mw), 499(m), 529(m), 604(mw), 633(w), 663(mw), 740(w), 775(m), 792(mw), 808(mw), 857(vs), 876(s), 925(m), 942(vs), 1003(mw), 1042(mw), 1056(mw), 1075(mw), 1095(m), 1123(mw), 1142(m), 1160(mw), 1192(w), 1258(w), 1282(m), 1293(m), 1317(s), 1341(s), 1353(m), 1401(vs), 1421(s), 1434(s), 1517(w), 1536(w), 1570(w), 1605(mw), 1624(mw), 1641(w), 1683(w)	2(e)
L-Arginine	376(w), 410(w), 490(mw), 551(m), 577(m), 613(m), 849(m), 879(mw), 922(m), 982(vs), 1036(mw), 1067(s), 1100(m), 1122(m), 1189(m), 1264(mw), 1298(m), 1330(m), 1377(mw), 1436(s), 1475(m), 1713(w)	2(f)
L-Phenylalanine	361(w), 468(m), 527(w), 605(w), 622(m), 747(mw), 832(m), 852(m), 951(w), 1004(vs), 1037(m), 1157(w), 1183(w), 1216(m), 1308(w), 1336(w), 1353(w), 1447(w), 1586(m), 1602(m)	3(a)
L-Tyrosine	379(w), 431(m), 491(w), 527(w), 641(m), 714(w), 797(mw), 828(vs), 845(m), 983(m), 1044(w), 1177(ms), 1200(m), 1214(m), 1247(w), 1265(m), 1326(m)	3(b)
L-Tryptophan	393(w), 425(w), 456(w), 498(m), 509(m), 534(m), 548(w), 574(m), 596(m), 626(m), 683(w), 706(m), 741(m), 755(vs), 766(m), 778(m), 802(w), 840(m), 848(m), 865(m), 874(s), 988(w), 1009(vs), 1046(w), 1076(w), 1103(w), 1118(m), 1160(w), 1207(w), 1231(m), 1253(w), 1278(w), 1309(w), 1314(m), 1328(m), 1338(s), 1358(s), 1423(s), 1450(m), 1457(m), 1486(m), 1556(s), 1576(m), 1616(m)	3(c)
L-Histidine	404(m), 422(mw,sh), 540(mw), 623(mw), 656(m), 680(w), 731(mw), 784(mw), 804(m), 824(mw), 852(m), 918(m), 929(mw,sh), 963(m), 976(m), 1061(m), 1087(s), 1111(m), 1140(mw), 1174(m), 1224(m), 1250(m), 1271(s), 1317(vs), 1336(m), 1347(m), 1407(m), 1430(m), 1476(mw), 1498(m), 1538(w), 1571(m), 1608(w), 1639(w)	3(d)

(continued overleaf)

Table 2. (Continued)

Product	Raman bands (in cm ⁻¹)	Fig
L-Proline	374(mw), 452(vs), 563(w), 577(w), 642(m), 681(mw), 774(w), 793(mw), 834(vs), 842(vs,sh), 850(s,sh), 877(m), 899(vs), 916(vs), 930(m), 947(m), 951(m,sh), 987(mw,sh), 994(m), 1035(m), 1045(m), 1083(m), 1166(mw), 1175(mw), 1194(mw), 1216(w), 1240(m), 1267(m), 1286(m), 1317(mw), 1333(mw), 1350(mw), 1357(mw), 1378(m), 1389(m), 1410(mw), 1418(mw), 1434(w,sh), 1443(mw,sh), 1454(s), 1479(mw), 1547(w), 1605(w)	3(e)
Lauric acid	464(w), 631(w), 664(w), 891(m), 906(m), 1018(w), 1038(w), 1061(s), 1078(m,sh), 1084(m), 1127(vs), 1178(w), 1191(w), 1272(w,sh), 1296(vs), 1369(w), 1408(m), 1446(vs), 1459(s), 1490(s), 1497(w), 1651(w)	4(a)
Myristic acid	414(m), 421(w,sh), 670(w), 893(m), 908(m), 986(w), 1039(w), 1063(vs), 1092(m), 1128(vs), 1175(w), 1296(vs), 1371(w), 1412(m), 1422(s), 1438(s), 1454(m), 1507(w), 1627(w), 1650(w)	4(b)
Palmitic acid	375(m), 433(w), 670(w), 893(m), 909(m), 1063(vs), 1099(m), 1129(vs), 1174(w), 1296(vs), 1371(w), 1421(s), 1438(vs), 1455(m), 1465(s), 1481(w), 1627(w)	4(c)
Stearic acid	818(w), 848(m), 866(m), 890(w), 909(w), 1062(s), 1100(m), 1129(s), 1173(m), 1187(w), 1206(w), 1296(s), 1371(w), 1408(m), 1418(m), 1441(vs), 1463(s), 1474(m), 1498(w), 1648(w)	4(d)
12-Methyl-tetradecanoic acid (15Aiso)	433(w), 669(w), 772(w), 818(m), 840(w), 905(m), 983(w), 990(w), 1018(w), 1040(w), 1063(s), 1098(s), 1130(w,sh), 1141(s), 1171(w), 1265(w), 1296(s), 1302(m), 1344(w), 1371(w), 1380(w), 1412(m), 1440(s), 1447(s), 1459(s), 1472(s), 1505(w), 1649(w)	5(a)
13-Methylmyristic acid (15iso)	435(w), 822(m), 907(m), 917(w), 936(w), 1061(s), 1098(m), 1134(s), 1170(w), 1272(w), 1296(vs), 1371(w), 1410(w), 1447(vs), 1469(s), 1492(w), 1649(w)	5(b)
14-Methyl-pentadecanoic acid (16iso)	418(m), 434(w), 666(w), 815(m), 831(w), 904(m), 918(w), 940(w), 952(w), 1048(w,sh), 1061(s), 1102(m), 1135(s), 1170(w), 1244(w), 1270(w,sh), 1284(m), 1296(s), 1337(w), 1370(w), 1412(m), 1445(s), 1465(s), 1651(w)	5(c)
14-Methylhexadecanoic acid (17Aiso)	372(m), 452(w), 667(w), 771(m), 788(w), 815(m), 834(m), 906(m), 969(w), 983(w), 995(w), 1006(w), 1031(m), 1062(vs), 1104(vs), 1140(s), 1170(w), 1186(w), 1253(w), 1296(s), 1344(m), 1370(m), 1411(m), 1441(vs), 1465(w), 1479(m), 1649(m)	5(d)
15-Methylpalmitic acid (17iso)	405(m), 416(m), 467(w), 516(w), 666(w), 801(w), 820(s), 846(w), 905(m), 910(m), 930(w), 952(w), 1010(w), 1050(w,sh), 1061(vs), 1103(s), 1135(vs), 1171(m), 1188(w), 1259(w), 1296(vs), 1306(s), 1323(w), 1340(w), 1369(w), 1386(w), 1410(m), 1448(vs), 1464(vs), 1501(w), 1648(w)	5(e)
Oleic acid	602(w), 725(w), 845(m,sh), 856(m), 866(m), 890(m), 903(m), 971(m), 1023(m,sh), 1035(m,sh), 1065(m), 1080(m), 1118(m), 1265(m), 1301(s), 1416(m,sh), 1440(vs), 1655(s)	6(a)
Vaccenic acid	817(w), 863(m), 889(m), 971(w), 1018(w), 1034(w), 1065(m), 1078(m), 1091(m), 1112(w,sh), 1264(m), 1302(s), 1416(w,sh), 1439(vs), 1655(s)	6(b)
Glycerol	392(s), 416(m), 485(s), 548(mw), 675(m), 820(s), 850(vs), 923(m), 976(mw), 1055(vs), 1110(vs), 1257(m, br), 1315(m), 1465(vs)	6(c)
Triolein	601(w), 725(w), 852(m), 869(m), 881(m), 971(mw), 1065(m), 1081(m), 1118(mw), 1266(m), 1301(s), 1440(vs), 1655(s), 1743(mw)	6(d)
Trilinolein	841(mw), 868(m), 912(mw), 972(m), 1075(m), 1108(mw), 1264(m), 1302(m), 1440(m), 1654(vs), 1743(w)	6(e)
Trilinolenin	866(m), 969(mw), 1024(mw), 1081(mw,br), 1160(mw), 1266(m), 1302(m), 1441(m), 1640(sh), 1655(vs), 1741(w)	6(f)
β -D-glucose	405(s), 440(m), 542(s), 650(mw), 772(mw), 841(ms), 914(ms), 1002(mw), 1022(m), 1054(m), 1075(ms), 1120(ms), 1149(m), 1272(m), 1296(ms), 1459(m)	7(a)
Lactose	377(vs), 399(m), 421(mw), 446(mw), 477(s), 555(mw), 568(w), 633(mw), 648(w), 677(w), 700(w), 782(w), 851(m), 876(m), 900(mw), 916(ms), 953(mw), 1005(mw), 1021(ms), 1031(m), 1041(m), 1053(ms), 1087(ms), 1120(m), 1142(ms), 1168(w), 1222(w), 1261(m), 1296(mw), 1326(m), 1359(m), 1380(m), 1414(w), 1455(mw), 1470(w)	7(b)
Cellulose	380(m), 436(w), 458(m), 493(m), 520(mw), 577(mw), 896(m), 969(w), 998(w), 1046(mw), 1061(m), 1096(vs), 1120(s), 1147(m), 1266(mw), 1337(m), 1379(m), 1413(m), 1461(m)	7(c)
D-(+)-Dextrose (α -D-glucose)	407(s), 440(m), 541(s), 652(m), 772(m), 841(s), 914(m), 1022(m), 1054(m), 1075(m), 1108(m), 1149(m), 1272(m), 1330(m), 1346(m), 1459(m)	7(d)

Table 2. (Continued)

Product	Raman bands (in cm ⁻¹)	Fig
D-(+)-Trehalose	369(s), 407(vs), 430(m), 450(s), 523(vs), 540(s), 580(mw), 604(m), 671(m), 697(m), 732(w), 804(m), 838(vs), 912(vs), 927(w), 957(mw), 1018(m), 1061(m), 1080(m), 1102(s), 1120(vs), 1149(vs), 1211(w), 1220(w), 1240(mw), 1261(mw), 1273(mw), 1313(mw), 1330(m), 1358(vs), 1371(m), 1386(m), 1411(m), 1455(m), 1469(mw)	7(e)
Amylose	407(m), 442(m), 481(vs), 579(m), 712(w), 758(w), 854(m), 901(m), 936(ms), 944(m), 1045(m), 1083(ms), 1123(ms), 1208(w), 1262(mw), 1267(m), 1339(m), 1379(m), 1460(m)	7(f)
Amylopectin	409(m), 439(m), 477(vs), 515(mw,sh), 578(mw), 612(w), 717(w), 762(m), 769(w), 866(m), 905(sh), 940(m), 1051(m), 1082(m), 1109(m), 1127(m), 1264(m), 1339(m), 1378(m), 1460(m)	7(g)
D-(+)-Mannose	397(s), 426(m), 447(s), 473(ms), 495(s), 530(s), 576(m), 604(m), 621(m), 683(m), 830(s), 861(s), 882(s), 914(m), 934(m), 958(m), 1033(m), 1087(s), 1106(s), 1139(ms), 1239(m), 1263(m), 1352(m), 1369(m), 1459(m), 1481(m)	8(a)
D-(+)-Fucose	383(w), 440(vs) 529(m), 669(m), 700(w), 772(mw), 815(m), 879(m), 964(mw), 1032(mw), 1076(mw), 1088(mw), 1113(m), 1129(m), 1155(m), 1256(m), 1273(m), 1331(m), 1452(m)	8(b)
D-(+)-Arabinose	406(w), 427(w), 512(mw), 580(m), 613(m), 698(m), 843(vs), 896(mw), 936(mw), 994(m), 1052(m), 1095(m), 1138(mw), 1259(m), 1311(mw), 1356(w), 1375(w), 1476(m)	8(c)
D-(+)-Xylose	410(m), 428(s), 504(m), 526(s), 565(m), 608(m), 757(mw), 903(vs), 930(m), 1017(mw), 1086(s), 1115(s), 1148(mw), 1315(mw), 1340(m), 1374(m), 1398(m), 1468(mw)	8(d)
D-(+)-Fructose	401(w,sh), 420(s), 463(m), 526(m), 595(m), 626(vs), 818(s), 872(s), 925(w), 978(m), 1048(m), 1060(w), 1082(m), 1143(w), 1176(w), 1250(w), 1265(s), 1340(w), 1399(w), 1455(w), 1471(m)	8(e)
D-(+)-Galactosamine	378(m), 421(m), 462(m), 530(vs), 606(mw), 657(w), 704(w), 776(w), 822(w), 872(vs), 887(m), 938(m), 948(m), 999(mw), 1020(m), 1038(mw), 1062(m), 1094(m), 1121(m), 1143(m), 1153(m), 1238(m), 1267(vs), 1318(w), 1336(m), 1369(mw), 1386(m), 1452(mw), 1468(w), 1514(mw), 1588(w), 1616(w)	9(a)
N-Acetyl-D-glucosamine	416(m), 463(w), 481(m), 513(s), 527(m), 554(m), 578(s), 630(mw), 698(w), 790(m), 865(m), 916(mw), 929(m), 973(vs), 1002(m), 1020(m), 1039(m), 1091(m), 1126(s), 1151(m), 1206(w), 1257(m), 1279(m), 1292(mw), 1321(m), 1330(m), 1361(m), 1380(m), 1432(m), 1473(m), 1552(m), 1627(m)	9(b)
Chitin	366(m), 396(m), 460(m), 498(m), 711(mw), 895(m), 955(m), 1059(m), 1107(s), 1149(m), 1205(mw), 1262(m), 1328(m), 1371(m), 1414(m), 1449(mw), 1626(m), 1656(m)	9(c)
Citric acid	380(s), 420(m), 489(w), 513(m), 554(m), 597(m), 641(w), 685(m), 784(vs), 881(w), 903(m), 942(s), 1053(m), 1082(m), 1145(w), 1170(w), 1211(m), 1345(w), 1390(m), 1426(w,sh), 1434(m), 1465(w), 1633(w), 1691(s), 1732(m)	10(a)
Succinic acid	386(s), 582(w), 684(m), 936(vs), 1086(m), 1230(w), 1294(m), 1369(m), 1420(s), 1432(s), 1457(m,sh), 1655(m)	10(b)
Fumarate	768(m), 913(m), 982(m), 1293(m), 1430(vs), 1563(w), 1593(w), 1629(m), 1640(m), 1657(s)	10(c)
Malic acid	400(mw), 440(mw), 535(mw), 611(s), 660(w), 749(s), 882(mw), 911(mw), 964(vs), 1033(m), 1098(m), 1176(w), 1186(w), 1222(w), 1261(w), 1277(w), 1309(w), 1351(m), 1378(mw), 1423(m), 1447(mw), 1633(mw), 1675(mw)	10(d)
Pyruvate	393(w), 435(mw), 547(w), 636(m), 834(s), 980(mw), 1197(w), 1371(m), 1408(vs), 1430(w,sh), 1628(w), 1658(mw), 1700(vs)	11(a)
Phosphoenolpyruvate	391(w), 406(w), 431(w), 451(mw), 465(w), 535(w), 548(mw), 741(w), 756(mw), 787(vs), 846(m), 863(mw), 891(w), 966(mw), 973(mw), 1034(s), 1085(w), 1097(w), 1185(w), 1234(w), 1266(m), 1286(w), 1318(w), 1355(w), 1373(mw), 1397(mw), 1446(m), 1472(w), 1631(mw)	11(b)
Coenzyme A	525(m), 559(m), 633(w), 663(m), 722(vs), 787(m,sh), 803(m), 879(m), 927(w), 949(w), 1027(m), 1053(m), 1100(m), 1157(w), 1203(m), 1248(m), 1325(s), 1407(s), 1420(s,sh), 1461(m), 1508(m), 1557(m), 1608(w), 1654(w)	11(c)
Acetyl coenzyme A	496(m), 531(m), 564(m), 631(s), 683(w,sh), 724(vs), 799(m), 882(m), 926(w), 948(w), 1026(m), 1045(m,sh), 1094(m), 1121(m), 1207(m), 1243(m), 1306(s), 1333(s), 1375(m), 1408(m), 1462(m), 1508(m), 1559(w), 1576(m), 1658(w)	11(d)
Acetoacetate	401(m), 486(m), 498(m), 519(m), 551(m), 582(m), 640(s), 719(w), 761(s), 830(vs), 903(m), 928(s), 1005(s), 1025(m), 1087(w), 1158(m), 1171(m), 1191(m), 1219(w), 1283(m), 1326(m), 1362(m), 1399(vs), 1422(vs), 1444(vs), 1580(m), 1600(w), 1626(w), 1653(w), 1703(vs), 1715(s,sh)	11(e)

(continued overleaf)

Table 2. (Continued)

Product	Raman bands (in cm^{-1})	Fig
D-Fructose-6-phosphate	390(m), 534(m), 617(m), 716(w), 834(m), 973(s), 1071(m), 1127(m), 1229(w,sh), 1266(m), 1347(m), 1376(m,sh), 1458(m), 1724(w)	11(f)
β -Carotene	1008(m), 1156(vs), 1190(m), 1211(m), 1270(w), 1280(w,sh), 1353(w), 1394(w), 1448(w), 1515(vs)	12(a)
Ascorbic acid	448(m), 566(s), 588(m), 629(s), 696(m), 710(m), 820(s), 1025(m,sh), 1130(vs), 1256(s), 1296(m), 1319(s), 1498(m), 1653(vs), 1667(vs)	12(b)
Riboflavin	422(w), 451(w), 502(m), 531(w), 603(w), 677(w), 742(m), 785(m), 1158(m), 1184(m), 1226(s), 1253(w), 1345(vs), 1402(m), 1464(m), 1496(m), 1534(m), 1576(m), 1621(w), 1658(w), 1704(w)	12(c)
Glutathione	400(s), 446(w), 523(w), 550(m), 625(vs), 643(m), 660(s), 679(vs), 722(w), 776(s), 811(m), 828(m), 867(m), 885(s), 917(m), 931(s), 953(m), 972(m), 988(m), 1015(m), 1041(m), 1074(w), 1117(w), 1143(w), 1169(m), 1224(m), 1235(m), 1255(w), 1280(s), 1309(m), 1334(m), 1368(m), 1403(m), 1415(m), 1443(m), 1455(m), 1536(w), 1629(m), 1660(w), 1703(w)	12(d)

Abbreviations: w, weak; m, medium; s, strong; mw, medium weak; ms, medium strong; vs, very strong; sh, shoulder; br, broad. The reference spectra can be requested at <http://www.analchem.ugent.be/Raman>.

appear in the spectra of the branched fatty acids. The spectra of the iso-fatty acids (fatty acids branched at the last but one carbon atom) clearly show some differences compared to the spectra of the Aiso acids (fatty acids branched at the third carbon atom, counted from the end of the carbon chain). Firstly, in the spectra of the Aiso acids, the relative intensity of band at 1296 cm^{-1} is lower than in the spectra of the iso acids. Secondly, for the Aiso acids the three bands in the region $1050\text{--}1150\text{ cm}^{-1}$ have about the same intensity, but for the iso acids, the band at approximately 1090 cm^{-1} is smaller compared to the two surrounding bands. Thirdly, in the region of the $\rho(\text{CH}_2)$ rocking vibrations, the Aiso fatty acids show only one band at approximately 906 cm^{-1} , while the iso-fatty acids show three to four extra bands between 910 and 970 cm^{-1} .

The Raman spectra of two mono unsaturated fatty acids are shown in Fig. 6: oleic acid (6a) and *cis*-vaccenic acid (6b). These acids are fluids and so their Raman spectra show broader bands compared to the saturated fatty acids, although all bands are located around the same wavenumber. For the unsaturated fatty acids (6a, b), an additional band at 1655 cm^{-1} appears due to $\nu(\text{C}=\text{C})$ stretching vibrations. These acids differ only in the position of the double bond (appendix), so their Raman spectra are highly similar. Only in the fingerprint region ($600\text{--}1200\text{ cm}^{-1}$) some minor differences can be observed. The spectrum of triolein (glycerol esterified 3 times with oleic acid) shows a very similar Raman pattern compared to oleic acid. Indeed, no contribution of the glycerol spectrum (6c) is seen in the three fats (6d, e, f): the glycerol spectrum shows clear bands at 416 and 485 cm^{-1} , a region where no bands are present in the spectra of the fats. The most important difference between the fatty acids (6a, b) and the fats (6d, e, f) is the presence of a weak band at about 1743 cm^{-1} that can be attributed to $\nu(\text{C}=\text{O})$ stretching vibrations from the ester bonds between glycerol and the fatty acids. Triolein, trilinolein and trilinolenin consist of glycerol esterified with identical

fatty acids containing respectively one, two and three double bonds (see appendix). Consequently the intensity of the $\nu(\text{C}=\text{C})$ stretching band at 1655 cm^{-1} increases compared to the other Raman bands in the spectra 6d, e and f. Next to some minor changes in the fingerprint region, it is noteworthy that the relative intensity of the bands at 1265 and 1301 cm^{-1} , due to respectively $\delta(\text{=CH})$ deformations and $\delta(\text{CH}_2)_n$ deformations, changes as the fats contain more double bonds.

Saccharides

Figure 7 shows the spectra of β -D-glucose and D-(+)-dextrose (α -D-glucose) and some of their dimers and polymers. The spectra of β -D-glucose (7a) and D-(+)-dextrose (7d) are very similar, but can be distinguished from one another in the regions from 1000 to 1200 cm^{-1} and from 1300 to 1500 cm^{-1} , which contain bands of $\nu(\text{C}-\text{O})$ and $\nu(\text{C}-\text{C})$ stretches respectively $\delta(\text{CH}_2)$ and $\delta(\text{CH}_2\text{OH})$ deformations.¹⁶ The spectra of D-(+)-dextrose (7d) and its dimer D-(+)-trehalose (7e) are also very similar, although a clear shift of an intense band can be observed from 541 to 523 cm^{-1} . This band is associated with exocyclic deformations, while the band at approximately 405 cm^{-1} can be assigned to endocyclic deformations.¹⁷ In the spectra of β -D-glucose (7a) and lactose (7b), both the endocyclic and exocyclic deformation bands show large shifts from 405 to 377 cm^{-1} and from 542 to 477 cm^{-1} respectively. In the regions of $\delta(\text{COH})$, $\delta(\text{CCH})$ and $\delta(\text{OCH})$ side group deformations ($800\text{--}950\text{ cm}^{-1}$), of $\nu(\text{C}-\text{O})$ and $\nu(\text{C}-\text{C})$ stretches ($950\text{--}1200\text{ cm}^{-1}$) and of $\delta(\text{CH}_2)$ and $\delta(\text{CH}_2\text{OH})$ deformations ($1250\text{--}1500\text{ cm}^{-1}$), the spectrum of lactose shows more bands than the spectrum of β -D-glucose (7a). This is probably due to the fact that lactose consists of β -D-glucose and another monosaccharide, *viz.* β -D-galactose. In the spectra of cellulose (7c), amylose (7f) and amylopectin (7g), polymers of β -D-glucose and D-(+)-dextrose, the Raman bands are much broader than those in the spectra of the monomers. Despite the broadening of the Raman bands

in the spectra of amylose (7f) and amylopectin (7g), they resemble the spectrum of their monomer D-(+)-dextrose (7d) in the region 800–1500 cm⁻¹. In the region below 700 cm⁻¹, intense bands for endo and exocyclic deformations were observed in the spectrum of D-(+)-dextrose (7d), but only one intense band is present in the spectra of amylose (7f) and amylopectin (7g). The branching in the structure of amylopectin does not induce many changes in its Raman spectrum compared to that of amylose. The spectrum of cellulose (7c) does not resemble the spectrum of its monomer β -D-glucose (7a).

The spectra of five other monosaccharides are shown in Fig. 8. Except for D-(+)-mannose, they can also appear in a furanose structure instead of a pyranose structure. For D-(+)-mannose, we noticed sample inhomogeneity and so the average spectrum contains many small bands (8a) and is hard to compare with the spectra of the other monosaccharides. D-(+)-fucose (8b), D-(-)-arabinose (8c) and D-(+)-xylose (8d) differ only in a modification of the CH₂OH group, but their Raman spectra are clearly different from one another and from the spectrum of β -D-glucose (7a).

Figure 9 presents spectra of three nitrogen containing saccharides. For D-(+)-galactosamine (9a), many shifts can be observed compared to the spectrum of β -D-glucose (7a), which are caused by the amino group and the change of a stereocenter (appendix). The most prominent band, compared to other monosaccharides, is the intense band at 1267 cm⁻¹, which is situated in the region of ρ (NH₂) rocking and δ (NH₂) twisting vibrations.²² For N-acetyl-D-glucosamine (9b), the presence of the amide group induces a much more complex Raman spectrum compared to β -D-glucose (7a). The band at 1627 cm⁻¹ is characteristic for the ν (C=O) stretch of an acetyl amide.²³ The spectrum of chitin (9c) shows broader bands than the spectrum of its monomer N-acetyl-D-glucosamine (9b).

Primary metabolites

Figures 10 and 11 show Raman spectra of primary metabolites. Figure 10 contains the spectra of four intermediates from the citric acid cycle: citric acid (10a), succinic acid (10b), fumarate (10c) and malic acid (10d). In the spectra of the acids (10a, b, d), a strong band is present in the region between 930 and 970 cm⁻¹ that can be assigned to the δ (OH...O) out of plane wagging vibration of intermolecular hydrogen bonds. The spectrum of citric acid shows a intense band at 784 cm⁻¹, which is probably caused by a δ (C₃CO) deformation. For fumarate (disodium salt, the acid was not commercially available), the symmetric ν (CO₂⁻) stretch results in a strong band at 1430 cm⁻¹, while the C=C bond that is conjugated with the carbonyl groups gives rise to a ν (C=C) stretch band at 1657 cm⁻¹. Although succinic acid, fumarate and malic acid are subsequent intermediates in the citric acid cycle and have readily similar chemical structures (appendix), their Raman spectra can be easily distinguished from each other. From comparison with biological materials, we suspect that

Raman bands in the spectra of the pure acids better resemble the bands in biological spectra better than Raman bands in the spectra of the corresponding salts (such as disodium fumarate).

Figure 11 presents Raman spectra of six metabolites that are related to the citric acid cycle. The spectra of pyruvate (11a) and phosphoenolpyruvate (11b) are very different, because the Raman bands of different functional groups dominate them. For pyruvate (11a), the most intense bands at 1408 and 1700 cm⁻¹ can be attributed to respectively the symmetric and asymmetric ν (CO₂⁻) stretch. The symmetric and asymmetric ν (POC) stretches give rise to the intense bands at 787 and 1034 cm⁻¹ in the spectrum of phosphoenolpyruvate (11b). The spectra of coenzyme A (11c) and acetyl coenzyme A (11d) are similar and both contain a band at approximately 723 cm⁻¹, which is the most intense band in the spectrum of adenine (1a). The ν (C-S) stretching bands of coenzyme A (11c) shifts from 663 cm⁻¹ to 631 cm⁻¹ in the spectrum of acetyl coenzyme A (11d) because of the conversion of the thiol to a thioester. For acetoacetate (11e), analogous to pyruvate (11a), the bands for symmetric and asymmetric ν (CO₂⁻) stretches are observed in the region from 1395 until 1450 cm⁻¹ and 1700 until 1720 cm⁻¹. The bands in the spectrum of D-fructose-6-phosphate (11f), are broader than the bands in the spectrum of D-(-)-fructose (8e). The asymmetric ν (POC) stretch of the phosphate group causes the most intense band (973 cm⁻¹) as this functional group is highly polarizable.

Others

Figure 12 presents Raman spectra of four other products that are of importance for living organisms. The spectrum of β -carotene (12a) is characterized by two intense bands at 1156 and 1515 cm⁻¹, that can be assigned to a combination of a ν (C-C) stretch and ν (C-H) deformation on the one hand and ν (C=C) stretching on the other hand. The spectrum of ascorbic acid (12b) shows a broad band at 1653–1667 cm⁻¹ due to the ν (C=O) stretch of the lactone. In the spectrum of riboflavin (12c), a strong band can be observed at 1226 cm⁻¹ that can be assigned to an aromatic vibration of the polycyclic structure. In the spectrum of glutathione (12d), intense bands are related to the presence of the sulphur atom: (1) ν (C-S) stretching causes intense bands in the region 600–700 cm⁻¹ and (2) the band at 400 cm⁻¹ can be assigned to δ (C-S) deformations. Furthermore, the ν (C=O) stretching of amide and carboxylic groups gives rise to a broad band at 1630 cm⁻¹.

Examples of Raman spectra of biological materials

All these reference spectra can be used for the interpretation of Raman spectra of biological materials. To illustrate the applicability of these reference spectra, we selected two Raman spectra of biological materials. In Fig. 13, Raman spectra of fungal spores (13a) and bacteria (13b) are shown. There is a high similarity between the spore spectra of *Amanita phalloides* (13a) and the spectra of fatty acids and

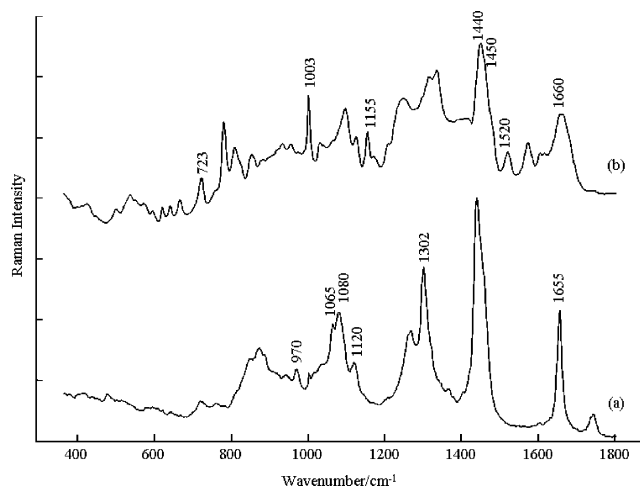


Figure 13. Raman spectra of biological materials: (a) spores of *Amanita phalloides* and (b) *Bacillus vallismortis* LMG 18725.

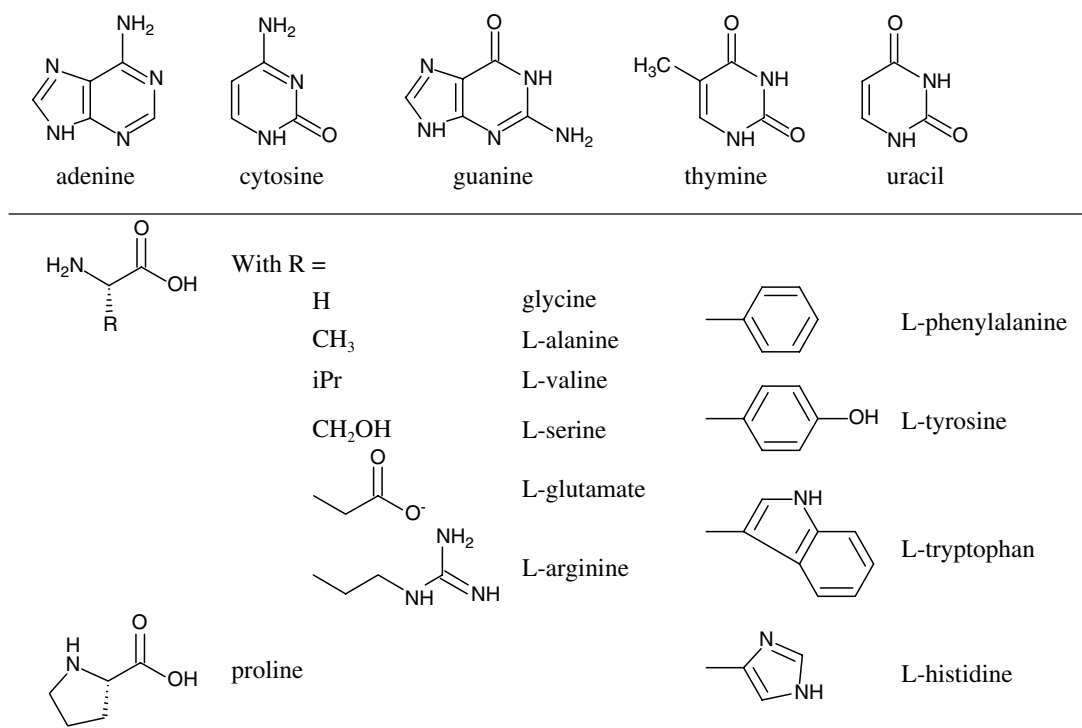
fats (Figs 4–6). Indeed, the typical bands for these molecules that were mentioned above appear the region from 1000 to 1700 cm^{-1} . Therefore, we can conclude that fatty acids and fats are the main components of these spores. The bands at 970, 1065, 1081, 1121, 1302, 1440 and 1656 cm^{-1} can be attributed to triolein and trilinolenin. Oleic acid bands can be observed at 1036, 1302 and 1440 cm^{-1} . The amyloidic ornamentation, consisting of amylopectin, is observed in this

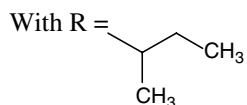
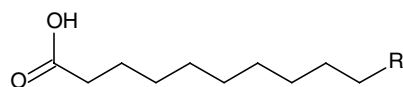
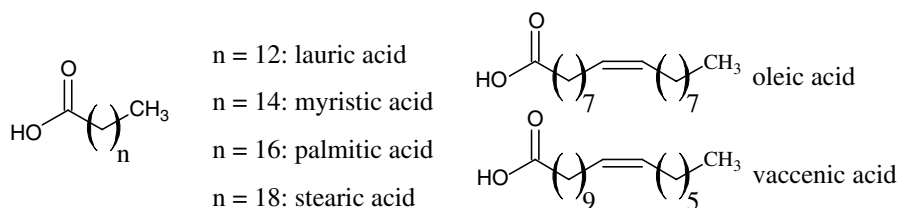
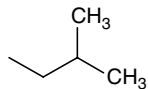
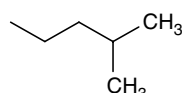
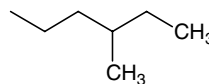
spectrum through the Raman bands at 477 and 864 cm^{-1} . In the spectrum of *Bacillus valismortis* LMG 18725 (13b), clear bands of fatty acids (1450(br), 1660(br) cm^{-1}), phenylalanine (1003 cm^{-1}), adenine (723 cm^{-1}) and carotenes (1155 and 1520 cm^{-1}) can be observed.

CONCLUSION

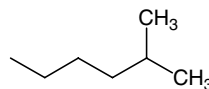
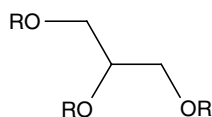
We presented Raman spectra of six groups of biomolecules. Most of these spectra can be easily distinguished from each other. The spectra of the DNA and RNA bases show almost no similarities so it is easy to distinguish between them and no characteristic bands are observed for either purine or pyrimidine bases. The spectra of all bases are characterized by intense ring breathing vibrations in the region from 600 until 800 cm^{-1} . The Raman bands of the functional groups present in the R side chain dominate the Raman spectra of amino acids. Only structurally very similar molecules, such as fatty acids mutually and saccharides mutually, give rise to similar Raman spectra. The spectra of fatty acids can be recognized by three intense bands in the region from 1000 until 1200 cm^{-1} and bands at about 1296 and 1440 cm^{-1} . For fats, the most prominent bands are those at approximately 1265, 1302, 1440 and 1655 cm^{-1} . Examples of Raman spectra of bacteria and fungal spores show that these reference spectra can be used to analyze complex biological Raman spectra.

APPENDIX

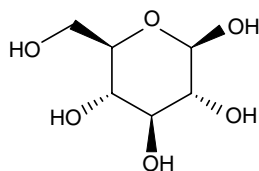
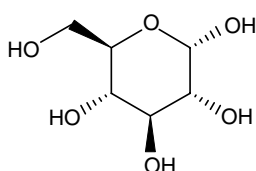


12-methyltetradecanoic
acid (15Aiso)13-methylmyristic
acid (15iso)14-methylpentadecanoic
acid (16iso)

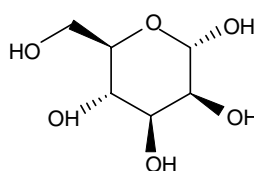
14-methylhexadecanoic
acid (17Aiso)

15-methylpalmitic acid
(17iso)

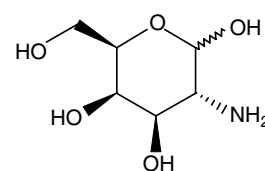
R = H: glycerol

$$-(\text{C}=\text{O})-(\text{CH}_2)_7-\text{CH}=\text{CH}-(\text{CH}_2)_7-\text{CH}_3: \text{triolein}$$
$$-(\text{C}=\text{O})-(\text{CH}_2)_7-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}-(\text{CH}_2)_4-\text{CH}_3: \text{trilinolein}$$
$$-(C=O)-(CH_2)_7-CH=CH-CH_2-CH=CH-CH_2-CH=CH-CH_2-CH_3$$
: trilinolenin β -D-glucose

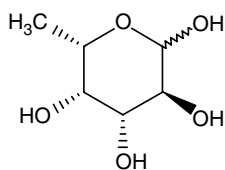
D-(+)-dextrose



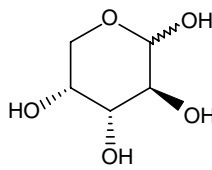
D-(+)-mannose



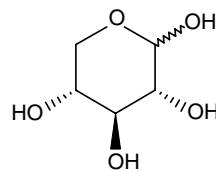
D-(+)-galactosamine



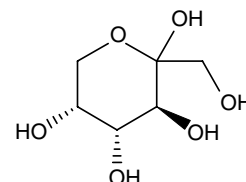
D-(+)-fucose



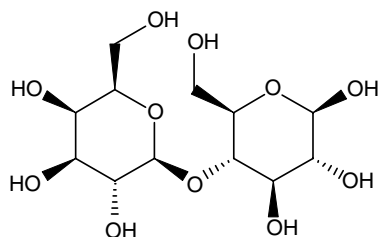
D-(-)-arabinose



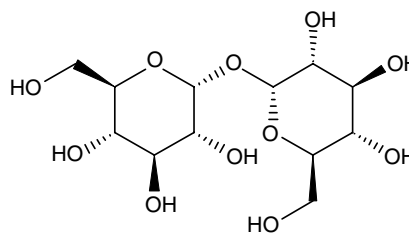
D-(+)-xylose



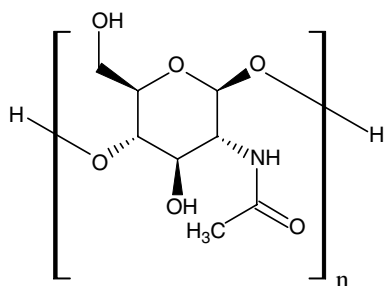
D-(-)-fructose



lactose

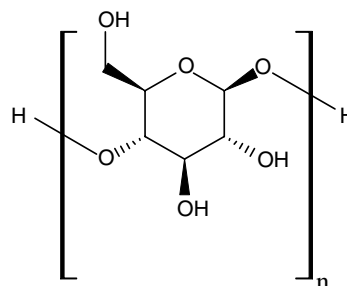


D-(+)-trehalose



$n = 1$: N-acetyl-D-glucosamine

$n \gg 1$: chitin



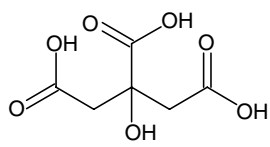
α -D-glucose:

linear polymer: amylose

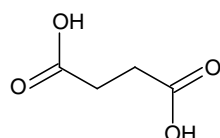
branched polymer: amylopectin

β -D-glucose:

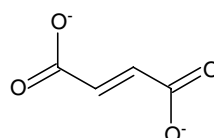
linear polymer: cellulose



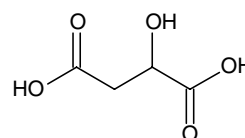
citric acid



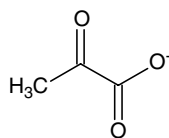
succinic acid



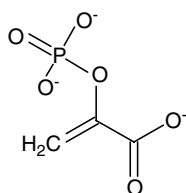
fumarate



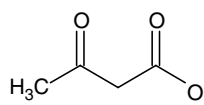
malic acid



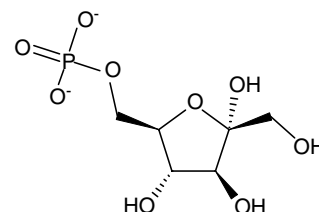
pyruvate



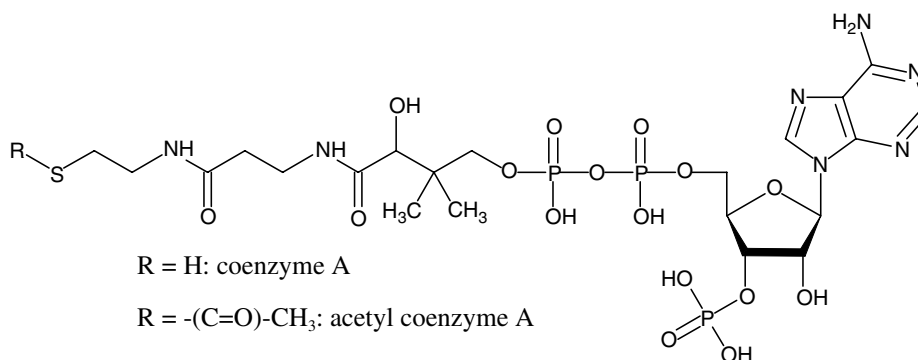
phosphoenolpyruvate



acetoacetate

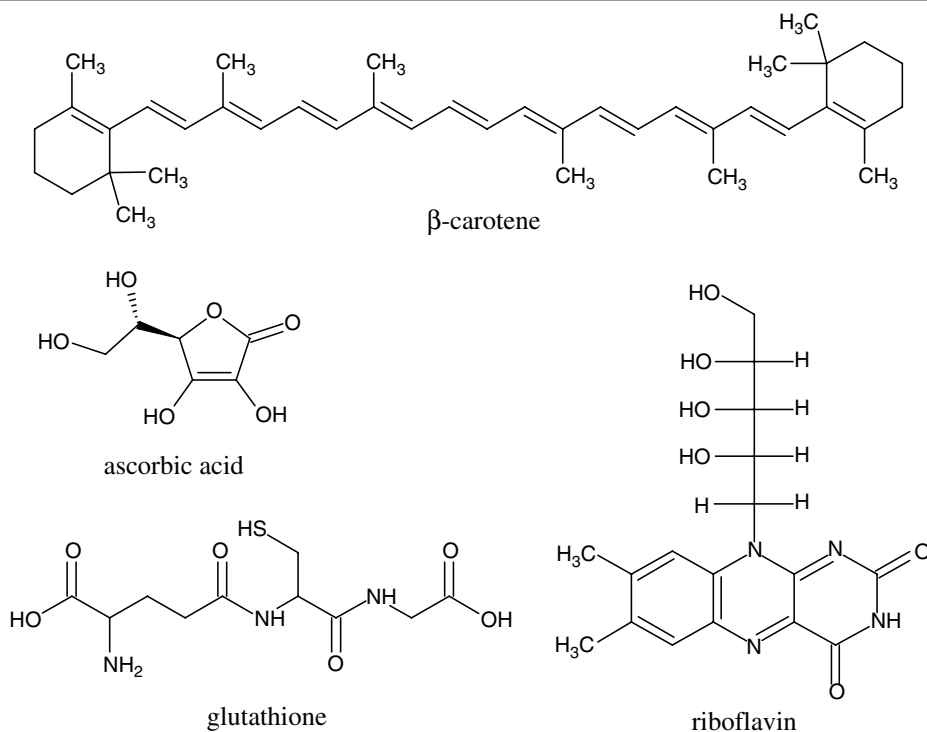


D-fructose-6-phosphate



$R = H$: coenzyme A

$R = -(C=O)-CH_3$: acetyl coenzyme A



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